



Final Report: Retrieval-Augmented Generation (RAG) System

Project: Generative AI Fundamentals Assignment

Architecture: LangChain + FAISS + Ollama (LLaMA 3)

Author: John Kiran Grandhi

1. Overview & System Architecture

The objective of this project was to develop a functional RAG application capable of ingesting specific research papers and providing context-aware answers through a conversational interface.

The Workflow:

1. **Ingestion:** Automated loading of 5 specific PDF documents via URL .
 2. **Processing:** Text extraction followed by recursive character chunking (Size: 500, Overlap: 50).
 3. **Embedding:** Conversion of text chunks into 384-dimensional vectors using the all-MiniLM-L6-v2 model.
 4. **Storage:** Implementation of a **FAISS** vector database for high-speed similarity searches.
 5. **Generation:** An **Ollama-based LLaMA 3** model generates responses by synthesizing retrieved context with the user query.
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2. Technical Implementation

Data Sourcing & Extraction

- **Libraries:** PyPDFLoader was utilized to handle the ingestion of the required research papers, including the seminal "Attention is All You Need" and "BERT" papers.
- **Chunking Strategy:** RecursiveCharacterTextSplitter was chosen to maintain semantic integrity, ensuring sentences aren't split mid-thought.

Conversational Memory

- **Mechanism:** A custom ConversationMemory class manages the state.
- **Constraint:** The system strictly maintains a sliding window of the **last 4 interactions**.

- **Prompt Engineering:** History is dynamically concatenated into the LLM prompt to ensure the bot "remembers" previous questions about Transformers or BERT.

3. Evaluation Framework

Following the assignment guidelines, a self-devised metric was implemented to score the 10 test questions.

Metric	Criteria	Score
Relevance	Keyword overlap between the query and the generated answer.	1 Pt
Contextual Awareness	Verification that the answer utilizes information from retrieved PDF chunks.	1 Pt
Response Quality	Length and clarity of the language (Character count > 80).	1 Pt
Basic Accuracy	Structural verification of meaningful statements.	1 Pt

4. Results & Discussion

Performance Analysis

- The system successfully handled complex queries regarding "Self-Attention" and "Masked Language Modeling" by retrieving highly relevant chunks from the provided PDFs.
- The conversational memory proved effective in maintaining context across the 10-question test suite.

Challenges Faced

- **PDF Formatting:** Handling multi-column research papers required careful chunking to avoid losing context between pages.

- **Inference Speed:** Running LLaMA 3 locally via Ollama requires significant RAM/GPU resources, which can impact response latency.

Potential Improvements

- **Advanced Evaluation:** Integrating the **RAGAS** framework for deeper faithfulness and relevancy scores.
- **Hybrid Search:** Combining vector similarity with keyword-based (BM25) search to improve retrieval of specific technical terms.

5. Conclusion

The implemented RAG pipeline meets all core requirements: PDF ingestion, vector database creation, LLM integration, and conversational memory. The system provides a robust foundation for interacting with complex technical literature through a natural language interface.